Security Dashboard (SIEM Lite)

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1. Tool Description
The Security Dashboard (SIEM Lite) project is a simplified implementation of a Security Information and Event Management (SIEM) system designed to monitor and analyze security logs generated during simulated cyber-attacks. The purpose of this tool is to demonstrate how real-world security monitoring systems detect suspicious activities such as brute-force attacks, port scanning, and exploitation attempts.
The lab environment was created using VMware virtualization software. Kali Linux was used as the attacker machine, while Metasploitable2 was used as the vulnerable target machine. The attacker machine generated simulated attacks, and the logs produced on the target system were collected and analyzed using Python scripts.
The main component of this project is the Python-based log analyzer tool. The tool reads system logs and identifies patterns that indicate malicious activity. For example, repeated failed login attempts are detected as brute-force attacks, scanning activity is identified as port scanning, and session creation logs are detected as exploitation attempts.
Additionally, a Flask-based web dashboard was developed to display detected alerts in an organized format. This dashboard provides a visual interface for viewing security alerts generated from log analysis.
The following tools were used in this project:

- Kali Linux (Attacker machine)
- Metasploitable2 (Target machine)
- VMware Workstation
- Python 3
- Flask Web Framework
- Nmap
- Hydra
- Metasploit Framework

This project demonstrates the basic working principles of SIEM systems and highlights the importance of log analysis in detecting security threats.
2. Test Results
The tool was tested using three different attack scenarios to validate its detection capability. Each test generated unique logs that were analyzed by the SIEM Lite system.
Test Case Results Table

Test Case	Attack Type	Tool Used	Input Description	Output Result
Test Case 1	Port Scan Detection	Nmap	Multiple port scan attempts on target system	Port scan alert generated
Test Case 2	Brute Force Detection	Hydra	Multiple failed SSH login attempts	Brute-force alert generated
Test Case 3	Exploit Detection	Metasploit	FTP exploit attempt on vulnerable service	Exploit alert generated

Test Case 1 — Port Scan Detection
Command Used:
nmap -A TARGET_IP
Description:
The Nmap tool was used to scan open ports on the Metasploitable system. This generated network scanning logs that were recorded in the system log files.
Result:
The SIEM Lite tool successfully detected scanning activity and generated a port scan alert.
Output Example:
ALERT: Port Scan Detected

Test Case 2 — Brute Force Attack Detection
Command Used:
hydra -l msfadmin -P rockyou.txt ssh://TARGET_IP
Description:
The Hydra tool was used to perform multiple SSH login attempts using password lists. This generated repeated failed login logs.
Result:
The system detected multiple failed login attempts and generated brute-force alerts.
Output Example:
ALERT: Brute Force Attempt Detected
Test Case 3 — Exploit Attempt Detection
Command Used:
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOST TARGET_IP
run
Description:
Metasploit was used to exploit a vulnerable FTP service available in Metasploitable.
Result:
The SIEM Lite tool detected exploit session logs and generated an exploit alert.
Output Example:
ALERT: Exploit Attempt Detected
3. Analysis of Findings
The results obtained from the testing phase demonstrate that the SIEM Lite tool effectively detects suspicious activities by analyzing system logs. Each test case produced different types of log entries, allowing the system to identify attack patterns using keyword matching techniques.
The port scan detection confirmed that unusual network scanning behavior can be recognized by analyzing system-level logs. Similarly, brute-force detection showed that repeated failed authentication attempts are reliable indicators of attack behavior. The exploit detection confirmed that unauthorized session creation logs indicate successful exploitation attempts.
This project highlights how log monitoring plays a critical role in identifying threats within a network environment. Even simple log analysis techniques can provide useful security insights and help administrators respond to potential attacks.
The dashboard interface also improves usability by presenting alerts in a structured manner. Visualizing alerts helps in quickly understanding the type and frequency of attacks.
4. Real Error Encountered
During the development of this project, an error occurred while transferring logs from the Metasploitable machine to the Kali machine using the SCP command.
Error Message:
ssh: connect to host 10.0.2.15 port 22: Connection refused
scp: Connection closed
Cause:
The SSH service was not running on the Metasploitable system, which prevented secure file transfer between the attacker and target machines.
Solution:
The issue was resolved by starting the SSH service manually using the following command:
sudo /etc/init.d/ssh start
After starting the SSH service, the SCP command worked successfully and logs were transferred correctly.
5. Ethical Considerations
All testing activities in this project were conducted within a controlled virtual lab environment using virtual machines. No real systems, networks, or users were targeted during testing. The tools used in this project were applied strictly for educational purposes and cybersecurity learning.
The use of vulnerable systems such as Metasploitable2 ensures that testing is performed safely without causing harm to real-world infrastructure. Ethical guidelines were followed throughout the project to ensure responsible use of cybersecurity tools.